Pujan Thing

Software Engineer | Backend & Machine Learning

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SUMMARY

Backend/ML engineer who built a fraud-detection system hitting 92% anomaly-detection accuracy on 500k+ transactions and a sentiment API serving 100+ concurrent requests at <200ms. Cut manual order-processing time 40% and data-entry effort 30% (15+ hrs/week saved) at Cyber Inquest. Currently building auth and AI-agent infrastructure at MyAgentsAI on FastAPI, Azure OpenAI, and PostgreSQL.

SKILLS

- **Backend:** Python, Django, DjangoDRF, FastAPI, REST APIs, GraphQL, PostgreSQL, MySQL, Microservices
- **ML & Data:** TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, PySpark
- **Cloud & DevOps:** Azure, AWS, Docker, CI/CD, GitHub Actions
- **Also:** React, Next.js, TypeScript, Firebase, Redis, Celery

EXPERIENCE

Software Engineer — **MyAgentsAI 10/2025 – 03/2026**

London, UK

- Owned end-to-end delivery of the platform's secure credential and provider login/signup system (NextAuth.js) as sole engineer, from design through production launch.
- Migrated the platform's AI agent infrastructure from OpenAI to Azure OpenAI, re-architecting the integration layer end-to-end.
- Rebuilt middleware.ts to standardize error-handling and redirect logic across the application's request pipeline.
- Shipped backend improvements to the FastAPI/Azure PostgreSQL stack every sprint since joining, as a core contributor to the platform team.
- Replaced manual release steps with a fully automated CI/CD pipeline (GitHub Actions, Docker, Azure ACR/App Service) for Next.js and FastAPI deployments.

Independent Machine Learning & Backend Projects — **Self-Directed 11/2022 – 10/2025**

- Achieved 92% anomaly-detection accuracy on a 500k+ transaction dataset by building a fraud detection system with TensorFlow/PyTorch autoencoders.
- Delivered >90% classification accuracy at 100+ concurrent requests and <200ms latency by building a real-time BERT + FastAPI sentiment analysis service.
- Shipped 5 production-grade ML/backend projects using TensorFlow, PyTorch, and FastAPI, publishing all code open-source on GitHub.

Software Engineer — **Cyber Inquest Pvt Ltd 01/2021 – 10/2022**

- Cut manual order-processing time by 40% and improved inventory accuracy by building a B2B order-management and store-stock monitoring platform (Python, MySQL, JavaScript, Flutter).
- Cut PDF data-entry effort by 30%, saving 15+ hours per week, by automating PDF data-extraction pipelines in Python.
- Delivered scalable Django REST APIs and background workers powering core user-facing functions, integrating seamlessly with the frontend.
- Ramped 3 interns to independent contribution by mentoring them on Django and engineering best practices.

EDUCATION

Bachelor of Engineering — **National College of Engineering**

- Major in Computer Engineering (Full Scholarship)

CERTIFICATIONS

- Google TensorFlow Developer Certificate (in progress)
- Coursera / DeepLearning.AI Machine Learning Specialization
- AWS Certified Developer – Associate (targeted)

PROJECTS

- **MediAlertHub** — FastAPI, GraphQL, React, Docker, LangChain: Healthcare logistics platform for real-time disease reporting and outbreak analysis; built scalable microservices with FastAPI, integrated GraphQL + REST APIs, and implemented a RAG pipeline with LangChain + FAISS for data summarization. Link
- **Quantum Harmonic Oscillator** (01/2025): Visualization of quantum mechanics on a harmonic oscillator using Jupyter Notebook, SciPy and NumPy. Link
- **Fraud Detection** (11/2024): Detected fraudulent financial transactions using TensorFlow/PyTorch autoencoders, achieving 92% accuracy on a 500k+ transaction dataset. Link
- **Sentiment Analysis** (09/2024): Real-time sentiment analysis web app using BERT and FastAPI, serving 100+ concurrent users with <200ms response time. Link
- **Goal Prediction** (08/2024): Revamped a Google Cloud tutorial with PySpark and TensorFlow to analyze 50k+ football shots for distance, angle, and goal prediction. Link
- **Algorithm Visualizer** (11/2023): Game visualizing the A* pathfinding algorithm using Pygame. Link
- **Major Project** (03/2021): Digitized traditional educational systems by developing a student learning platform using Python/Flask and machine learning.

OTHERS

- Certificate of Appreciation: Organizing Committee, Tech event YATRA, National College of Engineering, Nepal (02/2018)